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The deubiquitinase Ubp3/Usp10 constrains glucose-mediated mitochondrial repression via phosphate budgeting

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Abstract

Many cells in high glucose repress mitochondrial respiration, as observed in the Crabtree and Warburg effects. Our understanding of biochemical constraints for mitochondrial activation is limited. Using a *Saccharomyces cerevisiae* screen, we identified the conserved deubiquitinase Ubp3 (Usp10), as necessary for mitochondrial repression. Ubp3 mutants have increased mitochondrial activity despite abundant glucose, along with decreased glycolytic enzymes, and a rewired glucose metabolic network with increased trehalose production. Utilizing *Δubp3* cells, along with orthogonal approaches, we establish that the high glycolytic flux in glucose continuously consumes free Pi. This restricts mitochondrial access to inorganic phosphate (Pi), and prevents mitochondrial activation. Contrastingly, rewired glucose metabolism with enhanced trehalose production and reduced GAPDH (as in *Δubp3* cells) restores Pi. This collectively results in increased mitochondrial Pi and derepression, while restricting mitochondrial Pi transport prevents activation. We therefore suggest that glycolytic-flux dependent intracellular Pi budgeting is a key constraint for mitochondrial repression.

eLife assessment

This study explores the phenomenon of glucose-induced mitochondrial repression, known as the Crabtree effect, in cells cultured under high glucose conditions. The study uses a variety of well-designed assays to support the authors' hypothesis, shedding light on inorganic phosphate-mediated metabolic regulation. The analysis and conclusions could benefit from a more rigorous approach, given the limited replication on a single yeast strain background and inadequacies in the normalization methods, leaving the evidence in parts **incomplete**.

Introduction

Rapidly proliferating cells have substantial metabolic and energy demands in order to increase biomass (1, 2). This includes a high ATP demand, obtained from cytosolic glycolysis or mitochondrial oxidative phosphorylation (OXPHOS), to fuel multiple reactions (3). Interestingly, many rapidly proliferating cells preferentially rely on ATP from glycolysis/fermentation over mitochondrial respiration even in oxygen-replete conditions, and is the well-known Warburg effect (4, 5). Many such cells repress mitochondrial processes in high glucose, termed glucose-mediated mitochondrial repression or the Crabtree effect (6, 7). This is observed in some tumors (5), neutrophils (8), activated macrophages (9), stem cells (10–12), and famously *Saccharomyces cerevisiae* (7). Numerous studies have identified signaling programs or regulators of glucose-mediated mitochondrial repression. However, biochemical programs and regulatory processes in biology evolve around key biochemical constraints (13). The biochemical constraints for mitochondrial repression remain unresolved (14, 15).

There are two hypotheses on the biochemical principles driving mitochondrial repression. The first proposes direct roles for glycolytic intermediates in driving mitochondrial respiration, by inhibiting specific mitochondrial outputs (16, 17). The second hypothesizes that a competition between glycolytic and mitochondrial processes for mutually required metabolites/co-factors such as pyruvate, ADP or inorganic phosphate (Pi) could determine the extent of mitochondrial repression (14, 15, 18). These are not all mutually exclusive, and a combination of these factors might dictate mitochondrial repression in high glucose. However, any hierarchies of importance are unclear (19), and experimental data for the necessary constraints for mitochondrial repression remains incomplete.

One approach to resolve this question has been to identify regulators of metabolic state under high glucose. Post-translational modifications (PTMs) regulated by signaling systems can regulate mitochondrial repression (20–22). Ubiquitination is a PTM that regulates global proteostasis (23, 24), but the roles of ubiquitination-dependent processes in regulating mitochondrial repression are poorly explored. Ubiquitination itself is determined by the balance between ubiquitination, and deubiquitinase (DUB) dependent deubiquitination (25). Little is known about the roles of DUBs in regulating metabolic states, making the DUBs interesting candidate regulators of mitochondrial repression.

In this study, using an *S. cerevisiae* DUB knock-out library-based screen, we identified the evolutionarily conserved deubiquitinase Ubp3 (mammalian Usp10) as required for mitochondrial repression in high glucose. Loss of Ubp3 resulted in mitochondrial activation, along with a reduction in the glycolytic enzymes-phosphofructokinase 1 (Pfk1) and GAPDH (Tdh2 and Tdh3). This consequently reroutes glucose flux and increases trehalose biosynthesis. This metabolic rewiring increases Pi release from trehalose synthesis, and decreases Pi consumption in glycolysis, to cumulatively increase Pi pools. Using *ubp3Δ* cells along with independent analysis of wild-type cells, and isolated mitochondrial fractions, we establish that glycolytic-flux dependent Pi allocations to mitochondria determines mitochondrial activity. Through these data, we propose how intracellular Pi balance as controlled by glycolytic flux, is a biochemical constraint for mitochondrial repression.

Results

A deubiquitinase deletion screen identifies Ubp3 as a regulator of glucose-mediated mitochondrial repression

In cells such as *S. cerevisiae*, high glucose represses mitochondrial activity as well as OXPHOS-dependent ATP synthesis (7, 26) (illustrated in **Figure 1A**). Our initial objective was to identify proteostasis regulators of glucose-mediated mitochondrial repression. We generated and used a deubiquitinase (DUB) deletion strain library of *S. cerevisiae* (Figure S1A), to unbiasedly identify regulators of mitochondrial repression by measuring the fluorescence intensity of a potentiometric dye Mitotracker CMXRos (illustrated in **Figure 1B**). Using this screen, we identified DUB mutants with altered mitochondrial membrane potential (**Figure 1C**, Figure S1A). Note: WT cells in a respiratory medium (2% ethanol) were used as a control to estimate maximum mitotracker fluorescence intensity (Figure S1B).

A prominent 'hit' was the evolutionarily conserved deubiquitinase Ubp3 (**Figure 1C**), (homologous to mammalian Usp10) (Figure S1C). Due to its high degree of conservation across eukaryotes (Figure S1C) as well as putative roles in metabolism or mitochondrial function (27–29), we focused our further attention on this DUB. Cells lacking Ubp3 showed an ~1.5-fold increase in mitotracker fluorescence (**Figure 1C**, Figure S1A). Cells with catalyticaly inactive Ubp3 (Ubp3^{C469A}), showed increased mitochondrial potential comparable to *ubp3Δ* (**Figure 1D**). This data confirmed that Ubp3 catalytic activity of is required to fully repress mitochondrial activity under high glucose. The catalytic site mutation did not affect steady-state Ubp3 levels (Figure S1D).

Next, to assess the requirement of Ubp3 for mitochondrial function, we quantified the electron transport chain (ETC) complex IV subunit Cox2 (30)(**Figure 1E**). *ubp3Δ* had higher Cox2 than WT (**Figure 1E**). There was no increase in mitochondrial outer membrane protein Tom70 (Figure S1E), suggesting that the increased Cox2 is not merely because of higher total mitochondrial content. We next measured the basal oxygen consumption rate (OCR) of *ubp3Δ*, and basal OCR was higher in *ubp3Δ*, indicating higher respiration (**Figure 1F**).

Next, we asked if mitochondrial ATP synthesis was higher in *ubp3Δ*. The total ATP levels in WT and *ubp3Δ* were comparable (**Figure 1G**). However, upon treatment with the ETC complex IV inhibitor sodium azide, ATP levels in WT were higher than *ubp3Δ*, contributing to ~60% of the total ATP (**Figure 1H**). In contrast, the ATP levels in *ubp3Δ* after sodium azide treatment were ~ 40% of the total ATP (**Figure 1H**). These data suggest a higher contribution of mitochondrial ATP synthesis towards the total ATP pool in *ubp3Δ*.

We next asked if *ubp3Δ* required higher mitochondrial activity for growth. In high glucose, WT show minimal growth inhibition in the presence of sodium azide, indicating lower reliance on mitochondrial function (**Figure 1I**). Contrastingly, *ubp3Δ* or Ubp3^{C469A} show a severe growth defect in the presence of sodium azide (**Figure 1I**). Deletion of ETC components Cox2, and the ATP synthase subunits Atp1 and Atp10 results in a severe growth defect in *ubp3Δ* compared to WT (Figure S1F). Together, these results indicate that the loss of Ubp3 makes cells dependent on mitochondrial ATP synthesis in high glucose.

Collectively these data show that in 2% glucose, *ubp3Δ* have high mitochondrial activity, respiration, and rely on this mitochondrial function for ATP production and growth. We therefore decided to use *ubp3Δ* cells to start delineating requirements for glucose-mediated mitochondrial repression.

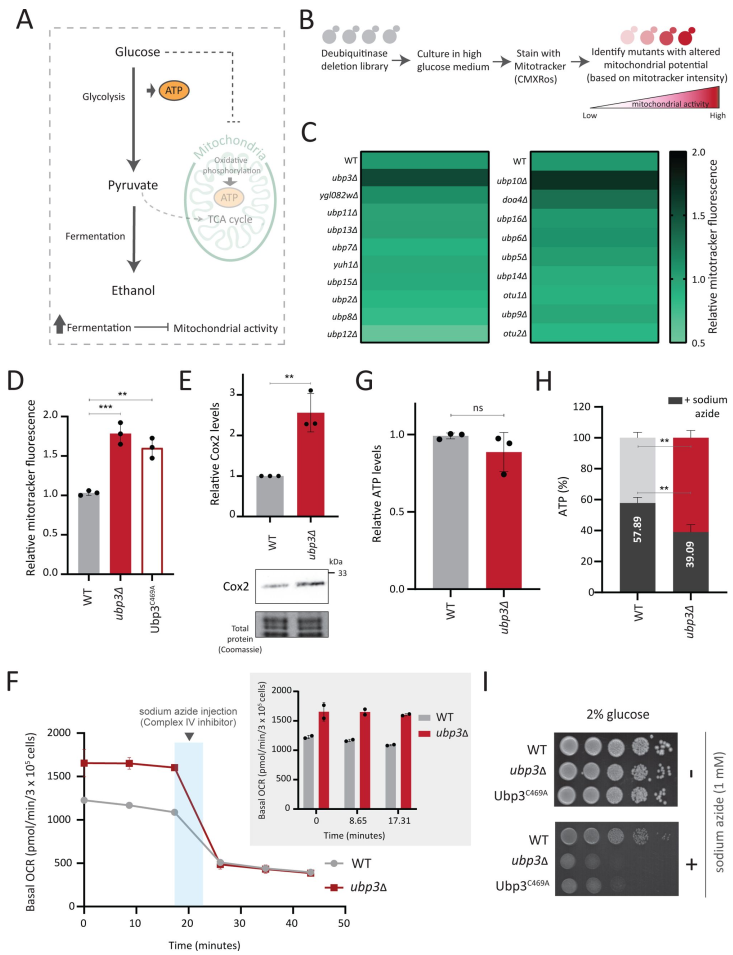

Figure 1

A deubiquitinase deletion screen identifies Ubp3 as a regulator of glucose-mediated mitochondrial repression

A) Schematic depicting glucose-mediated mitochondrial repression (Crabtree effect).
B) Schematic describing the screen with a yeast DUB KO library to identify regulators of Crabtree effect.
C) Identifying DUB knockouts with altered mitochondrial potential. Heat map shows relative mitochondrial membrane potential of 19 deubiquitinase deletions in high glucose, from 2 biological replicates. Also see figures S1A, S1B.
D) The deubiquitinase activity of Ubp3 and repression of mitochondrial membrane potential. WT, *ubp3Δ,* and Ubp3^{C469A} were grown inhigh glucose and relative mitochondrial membrane potential was measured. Data represent mean ± SD from three biological replicates (n=3). Also see figure S1D.
E) Effect of loss of Ubp3 on ETC complex IV subunit Cox2. WT and *ubp3Δ* were grown in high glucose, and Cox2 was measured (western blot using an anti-Cox2 antibody). A representative blot (out of 3 biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD.
F) Basal oxygen consumption rate (OCR) in high glucose in *ubp3Δ*. WT and *ubp3Δ* were grown in high glucose, and OCR was measured. Basal OCR corresponding to ~3 x 10^5 cells, from two independent experiments (n=2), normalized to the OD600 is shown. Bar graph representations are shown in the inset. Data represent mean ± SD.
G) Total ATP levels in *ubp3Δ* and WT. WT and *ubp3Δ* were grown in high glucose, and total ATP were measured. Data represent mean ± SD from three biological replicates (n=3).
H) Dependence of *ubp3Δ* on mitochondrial ATP. WT and *ubp3Δ* cells were grown in high glucose, and treated with 1 mM sodium azide for 45 minutes. Total ATP levels in sodium azide treated and untreated cells were measured. Data represent mean ± SD (n=3).
I) Requirement for mitochondrial respiration in high glucose in *ubp3Δ*. Serial dilution growth assay in high glucose in the presence/absence of sodium azide (1 mM) are shown. Also see figure S1F.
Data information: **p<0.01, ***p<0.001.

Key glycolytic enzymes decrease and glucose flux is rerouted in *ubp3Δ* cells

Glucose-6 phosphate (G6P) is the central node in glucose metabolism where carbon-allocations are made towards distinct metabolic arms, primarily: glycolysis, the pentose phosphate pathway (PPP), and trehalose biosynthesis (**Figure 2A**). We first compared amounts of two key ('rate-controling') glycolytic enzymes - Phosphofructokinase 1 (Pfk1), GAPDH isozymes (Tdh2, Tdh3) (3), along with the enolase isozymes (Eno1, Eno2) in WT and *ubp3Δ* cells. Pfk1, Tdh2, and Tdh3 substantially decreased in *ubp3Δ* (but not Eno1 and Eno2) (**Figure 2B**, S2A). Since deubiquitinases can control protein amounts by regulating proteasomal degradation, we asked if the decrease in Pfk1, Tdh2 and Tdh3 in *ubp3Δ* is due to proteasomal degradation. To test this, we measured the levels of these enzymes in *ubp3Δ* after treatement with proteasomal inhibitor MG132. We did not observe any rescue in the levels of these enzymes in MG132 treated samples, suggesting that the decreased levels of these enzymes were not due to increased proteasomal degradation (Figure S2B). To further understand if these enzyme transcripts are altered in *ubp3Δ*, we measured the mRNA levels of PFK1, TDH2 and TDH3 in WT, *ubp3Δ* and Ubp3^{C469A} cells. The transcripts of all the three genes in *ubp3Δ* and Ubp3^{C469A} cells decreased (Figure S2C), suggesting that Ubp3 regulates the transcripts of these glycolytic enzyme genes. The reduction in the Pfk and GAPDH enzyme amounts was intriguing, because the Pfk and GAPDH steps are critical in determining glycolytic flux (3, 31). A reduction in these enzymes could therefore decrease glycolytic flux, and would reroute glucose (G6P) allocations via mass action towards other branches of glucose metabolism, primarily the pentose phosphate pathway as well as trehalose biosynthesis (**Figure 2A**). To assess this, we first measured the steady-state levels of key glycolytic and pentose phosphate pathway (PPP) intermediates, and trehalose in WT or *ubp3Δ* using targeted LC-MS/MS. Glucose-6/fructose-6 phosphate (G6P/F6P) increased in *ubp3Δ* (**Figure 2C**). Concurrently, trehalose, and the PPP intermediates ribose 5-phosphate (R5P) and sedoheptulose 7-phosphate (S7P), increased in *ubp3Δ* (**Figure 2C**).

Since steady-state metabolite amounts cannot separate production from utilization, in order to unambiguously assess if glycolytic flux is reduced in *ubp3Δ* cells, we utilized a pulse labeling of $^{13}C_6$ glucose, following which the label incorporation into glycolytic and other intermediates was measured. Note that because glycolytic flux is very high in yeast, this experiment would require rapid pulsing and extraction of metabolites in order to stay in a linear range and avoid label saturation. We therefore established a very short time point of label-addition, quenching and metabolite extraction post ^{13}C glucose pulse. Since flux saturates/reaches steady-state in seconds, we first ensured that the label incorporation into individual metabolites after the ^{13}C glucose pulse was in the linear range, and for early glycolytic intermediates this was seconds after glucose addition. This new methodology is extensively described in materials and methods, with required controls shown in Figure S2E. WT and *ubp3Δ* cells were grown in high glucose, pulsed with $^{13}C_6$ glucose, and the relative ^{13}C label incorporation into glycolytic intermediates and trehalose were measured, as shown in the schematic (Figure S2D). In *ubp3Δ*, ^{13}C label incorporation into G6P/F6P as well as trehalose substantially increased (**Figure 2D**). Contrastingly, ^{13}C label incorporation into glycolytic intermediates F1,6BP, G3P, 3PG and PEP decreased, indicating decreased glycolytic flux (**Figure 2D**). We next measured ethanol concentrations and production rates, as an additional output of relative glycolytic rates. We observed decreased steady-state ethanol levels, as well as ethanol production rates in *ubp3Δ* (**Figure 2F**).

Collectively, these results reveal that that reduced Pfk1 and GAPDH (in *ubp3Δ*) decrease glucose flux via glycolysis, which results in rewired glucose flux towards trehalose biosynthesis and the PPP.

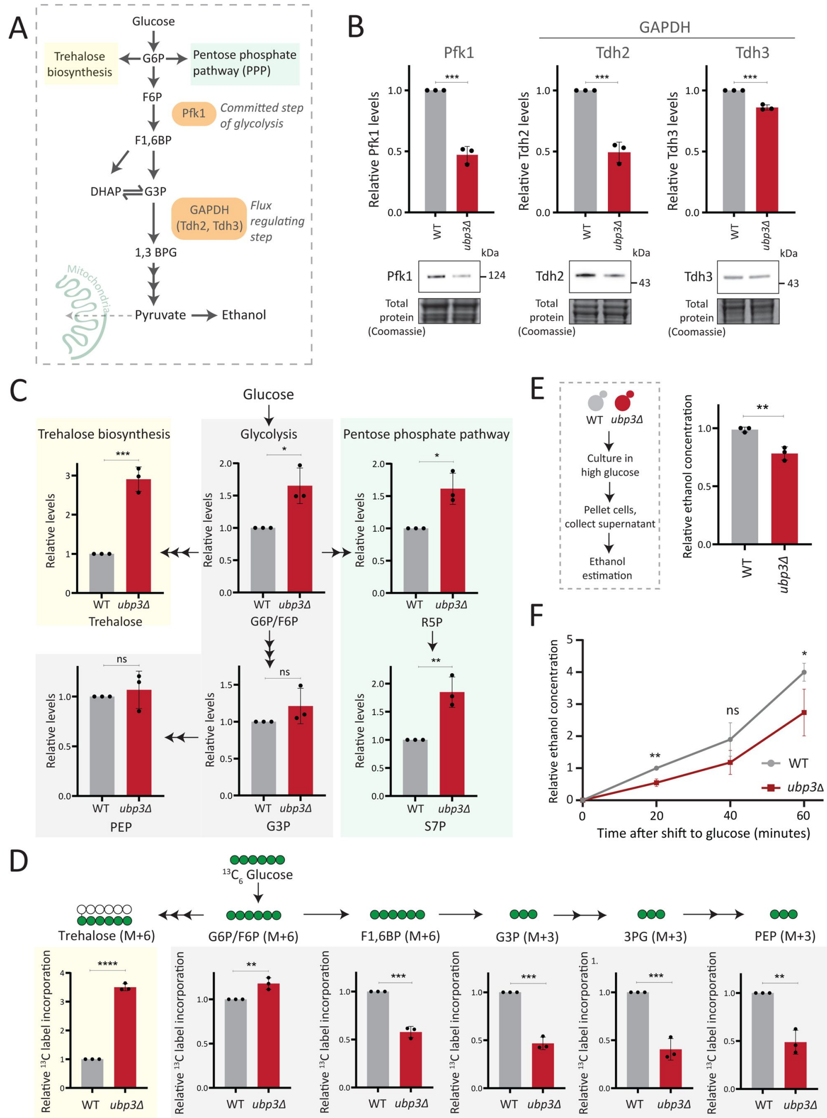

Figure 2

Key glycolytic enzymes decrease and glucose flux is rerouted in *ubp3Δ* cells

A) A schematic illustrating directions of glucose-6-phosphate (G6) flux in cells. Glucose is converted to G6P, a precursor for trehalose, the pentose phosphate pathway (PPP), and glycolysis.
B) Effect of loss of Ubp3 on key glycolytic enzymes. WT and *ubp3Δ* were grown in high glucose and the Pfk1, Tdh2, and Tdh3 levels were measured by western blot using an anti-FLAG antibody. A representative blot (out of three biological replicates, n=3) and their quantification are shown. Data represent mean ± SD. Also see figure S2A.
C) Steady-state metabolite amounts in WT and *ubp3Δ* in high glucose. Relative steady-state levels of trehalose, major glycolytic, and PPP intermediates were estimated in WT and *ubp3Δ*. Data represent mean ± SD from three biological replicates (n=3). Also see Appendix Table S3.
D) Relative glycolytic and trehalose synthesis flux in WT and *ubp3Δ*. Relative ^{13}C-label incorporation into trehalose and glycolytic intermediates, after a pulse of 1% $^{13}C_6$ glucose is shown. Data represent mean ± SD from three biological replicates (n=3). Also see Appendix Table S3, figures S2B, S2C.
E) Ethanol production in *ubp3Δ*. WT and *ubp3Δ* were grown in high glucose and ethanol in the media was measured. Data represent mean ± SD from three biological replicates (n=3).
F) Relative rate of ethanol production in WT vs *ubp3Δ*. WT and *ubp3Δ* were grown in high glucose (to OD_{600} ~ 0.6), equal numbers of cells were shifted to fresh medium (high glucose) and ethanol concentration in the medium was measured temporally. Data represent mean ± SD from three biological replicates (n=3)
Data information: *p<0.05, **p<0.01, ***p<0.001.

Rerouted glucose flux results in phosphate (Pi) accumulation

We therefore asked if the proteomic state, as observed in *ubp3Δ* cells, could provide clues to explain the coupling between mitochondrial derepression and rerouted glucose flux. A recent study by Isasa et al., had systematically quantified the changes in protein levels in *ubp3Δ* cells (28). We therefore reanalyzed this extensive dataset, looking for changes in proteins that would correlate with these metabolic processes. Notably, we observed increased levels of proteins of the mitochondrial ETC and respiration, and decreased amounts of glucose metabolizing enzymes in *ubp3Δ* (**Figure 3A**). Additionally, multiple proteins involved in regulating phosphate (Pi) homeostasis were decreased in *ubp3Δ* (**Figure 3A**) (28). We had recently uncovered a reciprocal coupling of Pi homeostasis with the different arms of glucose metabolism, particularly trehalose biosynthesis (32, 33), and therefore wondered if a glycolytic-flux dependent change in Pi homeostatsis had any role in mitochondrial respiration. Our hypothesis was refined based on the reasoning given below.

One explanation for mitochondrial repression can be an internal competition for shared metabolites/co-factors between (cytosolic) glycolytic and mitochondrial processes, that mitochondria might not be sufficiently able to access (14, 18, 19). In this context, a plausible role for inorganic phosphate (Pi) in regulating mitochondrial repression can be hypothesized. Cytosolic glycolysis requires rapid, high consumption of net Pi (19, 34, 35), and this could possibly limit the Pi that is continuously available for mitochondrial use (18, 36), thereby repressing mitochondria. Contextually, the balance between reactions releasing vs consuming Pi could explain changes in global Pi levels (33). A continuous hub of Pi consumption is glycolysis. Here, GAPDH catalyzes G3P to 1,3BPG, converting ADP to ATP, while concurrently consuming a molecule of Pi (35, 37). This Pi that goes into ATP will subsequently be used for nucleotide biosynthesis, polyphosphate biosynthesis and protein phosphorylation (33, 38). Therefore, we can surmise that in high glycolytic flux, the production of ATP, nucleotides and polyphosphates is concurrent with Pi consumption (37, 39, 40). Could this reaction therefore limit cytosolic Pi for the mitochondria (as illustrated in **Figure 3B**)? Notably, *ubp3Δ* have reduced GAPDH levels and decreased glycolytic flux. Second, trehalose synthesis is a Pi releasing reaction, and a major source of free Pi that is critical for Pi homeostasis (32, 35). Flux through this reaction is also substantially higher in *ubp3Δ*. Therefore, these cells might have increased Pi release (via trehalose), coupled with decreased Pi consumption (via decreased GAPDH). We therefore asked if total Pi increases in *ubp3Δ* (**Figure 3B**)?

To test this, we directly assessed total Pi levels in *ubp3Δ* and WT. *ubp3Δ* cells had higher Pi than WT, and this was comparable to Pi in WT grown in excess Pi (**Figure 3C**). Similarly, Pi amounts also increased in Ubp3^{C469A} (Figure S3A). Therefore, the loss of Ubp3 increases intracellular Pi levels. Next, we asked if *ubp3Δ* cells exhibit signatures of a 'high Pi' state. *S. cerevisiae* maintains internal Pi balance by controlling the expression of multiple genes collectively known as the Pho regulon (41). The Pho regulon is induced under Pi limitation, and repressed during Pi sufficiency (32, 41). We assessed two major Pho proteins (Pho84: a high-affinity membrane Pi transporter, and Pho12: an acid phosphatase) in WT and *ubp3Δ* in high glucose. *ubp3Δ* cells have lower amounts of Pho84 and Pho12 (**Figure 3D**). Further, upon shifting to low Pi for one hour, Pho84 and Pho12 increased in *ubp3Δ*. These data suggest that reduced Pho84 and Pho12 amounts in *ubp3Δ* are because of increased Pi, and not due to altered Pho regulon function itself (**Figure 3D**). These data clarify earlier observations from *ubp3Δ* which noted reduced Pho proteins (28). Therefore, *ubp3Δ* constitutively have higher Pi, and likely a consequent decrease in Pho proteins.

We next asked if the increased Pi in *ubp3Δ* is because of altered G6P allocations towards different end-points, particularly trehalose synthesis, which can be a major node of Pi restoration (32, 35). We assessed the contribution of increased trehalose synthesis towards the high Pi in *ubp3Δ*, by estimating Pi levels in the absence of trehalose 6-phosphate phosphatase (Tps2), which catalyzes the Pi-releasing step in trehalose synthesis. Notably, loss of Tps2 in *ubp3Δ* decreased Pi

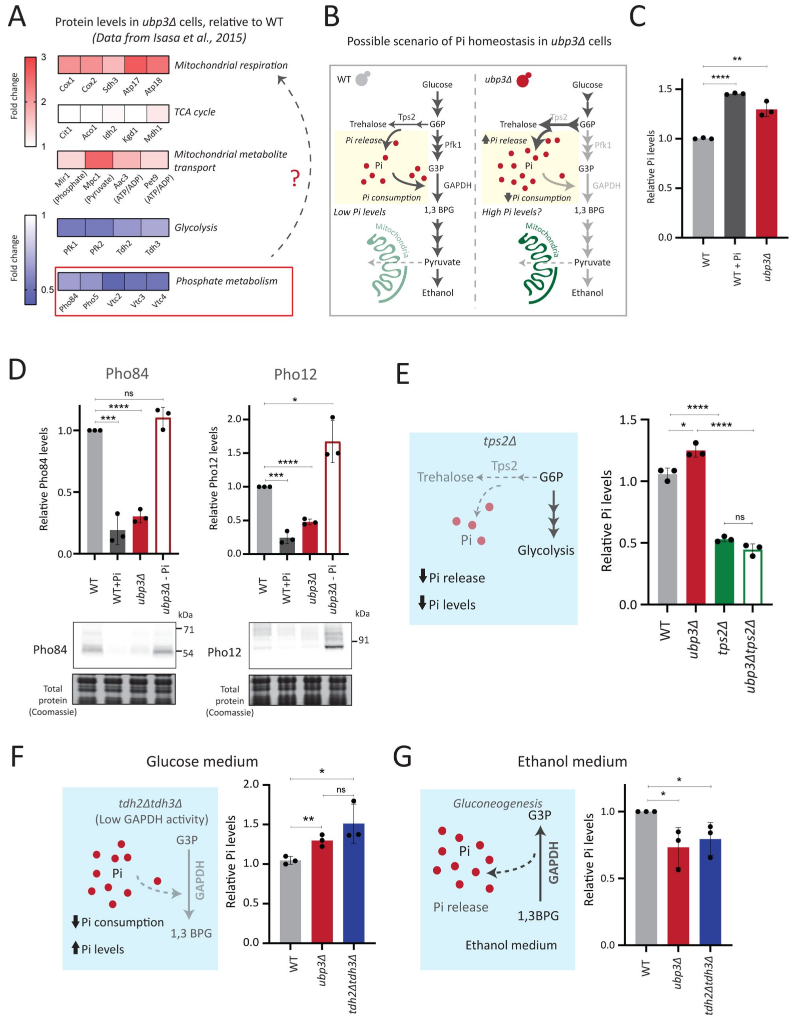

Figure 3

Rerouted glucose flux results in phosphate (Pi) accumulation

A) Changes in protein levels in *ubp3Δ* (dataset from Isasa et al., 2015). *ubp3Δ* cells have an increase in proteins involved in mitochondrial respiration and decrease in proteins involved in glucose and phosphate metabolism.
B) Schematic showing maintenance of Pi balance during glycolysis. Trehalose synthesis from G6P releases Pi, and the conversion of G3P to 1,3BPG by GAPDH consumes Pi. In *ubp3Δ*, trehalose biosynthesis (which releases Pi) increases. *ubp3Δ* have decreased GAPDH, which will decrease Pi consumption. This increase in Pi release along with decreased Pi consumption could increase cytosolic Pi.
C) Intracellular Pi levels in WT and *ubp3Δ*. WT and *ubp3Δ* were grown in high glucose and the total free phosphate (Pi) levels were estimated. WT in high Pi (2% glucose, 10mM Pi) was a positive control. Data represent mean ± SD from three biological replicates (n=3). Also see figure S3A.
D) Pho regulon responses in WT and *ubp3Δ*. Protein levels of Pho84-FLAG and Pho12-FLAG were compared between WT grown in high glucose and in high Pi, *ubp3Δ* in high glucose with or without a shift to a no-Pi medium for one hour, by western blot. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD.
E) Contribution of trehalose synthesis as a Pi source. WT, *tps2Δ*, *ubp3Δ,* and *ubp3Δtps2Δ* were grown in high glucose and the total Pi levels were estimated. Data represent mean ± SD from three biological replicates (n=3). Also see figure S3B.
F) Loss of GAPDH isozymes Tdh2 and Tdh3 and effect on Pi. WT, *ubp3Δ,* and *tdh2Δtdh3Δ* were grown in high glucose and total Pi was estimated. Data represent mean ± SD from three biological replicates (n=3).
G) Pi levels in *ubp3Δ* and *tdh2Δtdh3Δ* cells in ethanol medium. WT, *ubp3Δ,* and *tdh2Δtdh3Δ* cells were grown in ethanol medium and the total Pi levels were estimated. Data represent mean ± SD from three biological replicates (n=3).
Data information: *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

(**Figure 3E**). There was no additive difference in Pi between *tps2Δ* and *ubp3Δtps2Δ* (**Figure 3E**). As an added control, we assessed trehalose in WT and *ubp3Δ* in the absence of Tps2, and found no difference (Figure S3B). Therefore, increasing G6P flux towards trehalose biosynthesis is a major source of the increased Pi in *ubp3Δ*.

Since the major GAPDH isozymes, Tdh2 and Tdh3, are reduced in *ubp3Δ*, we directly asked if reducing GAPDH can decrease Pi consumption and increase Pi. To assess this, we generated *tdh2Δtdh3Δ* cells, which exhibit a growth defect, but are viable, permitting further analysis. *tdh2Δtdh3Δ* had higher Pi in high glucose (**Figure 3F**). Expectedly, we observed a decrease in ethanol in *tdh2Δtdh3Δ* (Figure S3C), along with an accumulation of F1,6 BP and G3P, and decreased 3PG and PEP (Figure S3D). However, G6P and trehalose levels between WT and *tdh2Δtdh3Δ* were comparable (Figure S3D, S3E). These data sugest that unlike in *ubp3Δ*, the increased Pi in *tdh2Δtdh3Δ* comes mainly from decreased Pi consumption (GAPDH step). To further assess the role of reduced glycolytic flux in increasing Pi (*ubp3Δ* and *tdh2Δtdh3Δ*), we measured the Pi in these cells growing in a gluconeogenic medium – 2%, ethanol. In this scenario, the GAPDH catalyzed reaction will be reversed, converting 1,3BPG to G3P, which should release and not consume Pi. Compared to WT, the Pi levels decreased in *ubp3Δ* and *tdh2Δtdh3Δ* (**Figure 3G**), suggesting that the changes in Pi in these mutants is driven by the relative change in Pi release vs consumption.

These results collectively indicate that the combined effect of increased Pi release coming from trehalose synthesis, and decreased Pi consumption from reduced GAPDH increase Pi levels in *ubp3Δ*.

Mitochondrial Pi availability correlates with mitochondrial activity in *ubp3Δ*

We therefore wondered if this observed Pi increase from the combined rewiring of glucose metabolism resulted in more Pi becoming accessible to the mitochondria. This would effectively result in Pi budgeting between cytosolic glycolysis and mitochondria based on the relative flux in different arms of glucose metabolism, by increasing Pi availability for the mitochondria (**Figure 4A**). To test if this were so, we first asked if the mitochondria in *ubp3Δ* have increased Pi. Mitochondria were isolated by immunoprecipitation from WT and *ubp3Δ*, the isolation efficiency was analyzed (Figure S4A), and relative Pi levels compared. Pi levels were normalised to Idh1 (isocitrate dehydrogenase) in isolated mitochondria, since Idh1 protein levels did not decrease in *ubp3Δ* (Figure S4B), and consistent with (28). Mitochondrial Pi was higher in *ubp3Δ* (**Figure 4B**). We next asked if the increased mitochondrial activity in *ubp3Δ* is a consequence of high Pi. If this were so, *tdh2Δtdh3Δ* should partially phenocopy *ubp3Δ* with respect to mitochondrial activity. Consistently, *tdh2Δtdh3Δ* have higher mitotracker intensity as well as Cox2 levels compared to WT (Figure S4C, **Figure 4C**). Also consistent with this, a high basal OCR in *tdh2Δtdh3Δ* was observed (**Figure 4D**), together indicating high mitochondrial activity in *tdh2Δtdh3Δ*, which is comparable to *ubp3Δ*.

We next asked if higher Pi is necessary to increase mitochondrial activity in *ubp3Δ* and *tdh2Δtdh3Δ*. Since glycolysis is defective in *ubp3Δ* and *tdh2Δtdh3Δ*, it is necessary to distinguish the effect of high Pi vs. only the effect of low glycolysis in activating mitochondria. Logically, if decreased glycolysis (independent of Pi) is sufficient to activate mitochondria, bringing down the Pi levels in *ubp3Δ* to that of WT should not affect mitochondrial activity. Notably, *ubp3Δ* grown in low (1 mM) Pi have Pi levels similar to WT in standard (normal Pi) medium (Figure S4D). *ubp3Δ* grown in low Pi also had decreased ethanol, suggesting reduced glycolysis (Figure S4E). Therefore, we used this condition to further understand the role of Pi in inducing mitochondrial activity. Mitotracker fluorescence decreased in both *ubp3Δ* and *tdh2Δtdh3Δ* in low Pi (Figure S4F). Note: basal mitotracker fluorescence in WT also decreases in low Pi, which is consistent with a required role of Pi for mitochondrial activity (Figure S4F). Similarly, Cox2 levels were reduced in both *ubp3Δ* and *tdh2Δtdh3Δ* (**Figure 4E**). Consistent with both reduced mitotracker intensity and Cox2

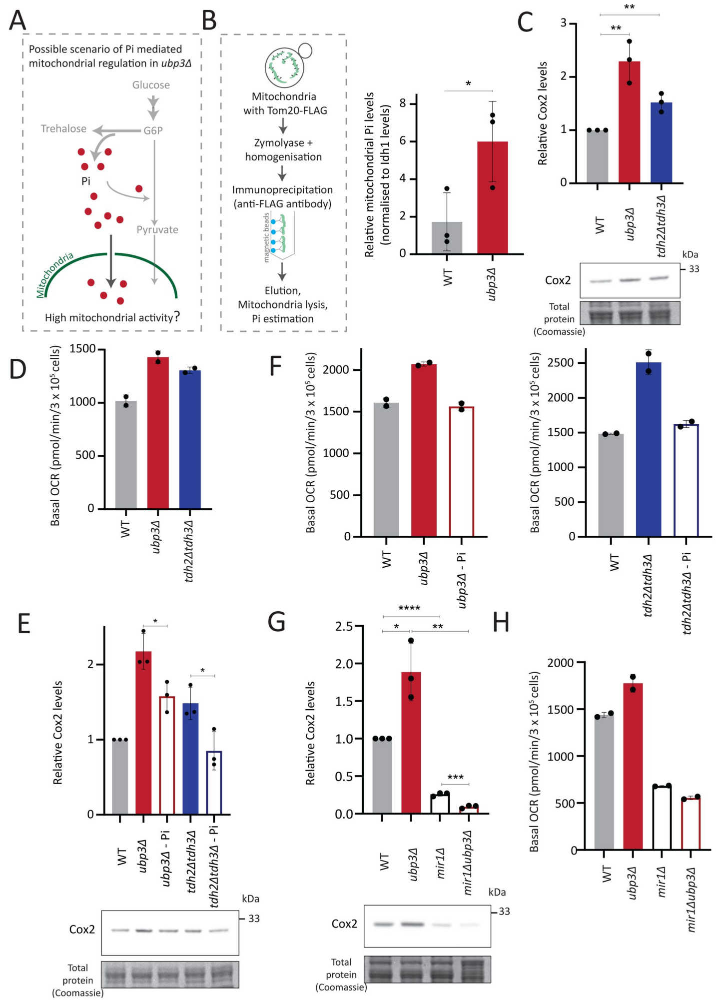

Figure 4

Mitochondrial Pi availability correlates with mitochondrial activity in *ubp3Δ*

A) A hypothetical mechanism of cytosolic free Pi controling mitochondrial activity by regulating mitochondrial Pi availability.
B) Mitochondrial Pi amounts in WT vs *ubp3Δ*. Mitochondria were isolated by immunoprecipitation from WT and *ubp3Δ* and mitochondrial Pi estimated. Mitochondrial Pi levels (normalised to Idh1) are shown. Data represent mean ± SD from three biological replicates (n=3). Also see figure S4A, B.
C) Cox2 protein in *tdh2Δtdh3Δ*. WT, *ubp3Δ,* and *tdh2Δtdh3Δ* were grown in high glucose and Cox2 protein was estimated. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD.
D) Basal OCR levels in *tdh2Δtdh3Δ*. WT, *ubp3Δ,* and *tdh2Δtdh3Δ* were grown in high glucose and basal OCR was measured from two independent experiments (n=2). Data represent mean ± SD. Also see figure S4C.
E) Comparative Pi amounts and Cox2 levels in *ubp3Δ*, *tdh2Δtdh3Δ,* WT cells. WT cells were grown in high glucose, *ubp3Δ* and *tdh2Δtdh3Δ* were grown in high glucose and low Pi, and Cox2 protein was estimated. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD. Also see figure S4D, S4F.
F) Pi amounts and basal OCR in *ubp3Δ* and *tdh2Δtdh3Δ* vs WT cells. WT cells were grown in high glucose, *ubp3Δ* and *tdh2Δtdh3Δ* were grown in high glucose and low Pi, and basal OCR was measured from two independent experiments (n=2). Data represent mean ± SD.
G) Effect of loss of mitochondrial Pi transporter Mir1 on Cox2protein. WT, *ubp3Δ*, *mir1Δ,* and *mir1Δubp3Δ* were grown in high glucose and Cox2 amounts compared. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD.
H) Relationship of mitochondrial Pi transport and basal OCR in WT vs *ubp3Δ*. WT, *ubp3Δ*, *mir1Δ,* and *mir1Δubp3Δ* cells were grown in high glucose and basal OCR was measured from two independent experiments (n=2). Data represent mean ± SD.
Data information: *p<0.05, **p<0.01, ****p<0.0001.

levels, basal OCR also decreased in both *ubp3Δ* and *tdh2Δtdh3Δ* in low Pi (**Figure 4F**). These data collectively suggest that high intracellular Pi is necessary to increase mitochondrial activity in *ubp3Δ* and *tdh2Δtdh3Δ*.

Next, we asked how mitochondrial Pi transport regulates mitochondrial activity. Mir1 and Pic2 are mitochondrial Pi transporters, with Mir1 being the major Pi transporter (42, 43). We first limited mitochondrial Pi availability in *ubp3Δ* by knocking-out *MIR1*. In *mir1Δ*, the increased Cox2 observed in *ubp3Δ* was no longer observed (**Figure 4G**). Consistent with this, we observed no further increase in the basal OCR in *mir1Δubp3Δ* compared to *mir1Δ* (**Figure 4H**). Furthermore, *mir1Δ* showed decreased Cox2 as well as basal OCR even in WT cells (**Figure 4G**, **Figure 4H**). These data together suggest that mitochondrial Pi transport is critical for increasing mitochondrial activity in *ubp3Δ*, and in maintaining basal mitochondrial activity even in high glucose.

As a control, no significant increase in Mir1 and Pic2 was observed in *ubp3Δ* (Figure S4G), suggesting that *ubp3Δ* do not increase mitochondrial Pi by merely increasing the Pi transporters, but rather by increasing available Pi pools.

Taken together, these data suggest the possibility that increased cytosolic Pi increases the mitochondrial Pi pool, which increases mitochondrial activity. Decreasing mitochondrial Pi by either reducing total Pi, or by reducing mitochondrial Pi transport decreases mitochondrial activity.

Mitochondrial Pi availability constrains mitochondrial activity under high glucose

So far, these data suggest that the cytosolic Pi available for the mitochondria can dermine the extent of mitochondrial activity. Therefore, we further investigated if mitochondrial Pi allocation was a necessary constraint for glucose-mediated mitochondrial repression.

To test this, we first asked how important mitochondrial Pi transport was to switch to increased respiration. In WT yeast, low glucose or glycolytic inhibition will result in increased respiration (22). What happens therefore if we restrict mitochondrial Pi in this context? For this, we measured the basal OCR in WT and *mir1Δ* after switching from high (2%) to low (0.1%) glucose. We observed a significant increase in the basal OCR in WT but not in *mir1Δ* (**Figure 5A**). The alternate scenario is after glycolytic inhibition. We assessed the role of mitochondrial Pi in this context, by inhibiting glycolysis using 2-deoxyglucose (2DG). WT, but not *mir1Δ*, increased their OCR (respiration) upon a 1-hour treatment with 2DG (**Figure 5B**). Consistent with this, mitotracker fluorescence increased with an increase in 2DG in WT, but not in *mir1Δ* (Figure S5A). We further asked if the mitochondrial Pi transporter itself glucose repressed, and therefore assessed Mir1 amounts in high and low glucose. Mir1 levels are higher upon a shift to low glucose, and in cells grown in 2% ethanol, suggesting that Mir1 is glucose repressed (**Figure 5C**, S5B). These data suggest that mitochondrial Pi transport is necessary for increasing mitochondrial activity after glucose derepression.

We next asked whether just adding external Pi was sufficient to increase mitochondrial activity, when cells are in high glucose. In medium supplemented with excess Pi, the internal Pi increases as seen earlier (**Figure 3C**). Therefore, a simplistic assumption would be that the addition of external Pi to cells in high glucose would also increase mitochondrial Pi. However, an alternate possibility presents itself whererin since glycolytic flux is already high in glucose, supplementing Pi will continue to fuel glycolysis. Indeed, this was originally obseved by Harden and Young in 1908, where adding Pi increased fermentation (44). In such a scenario, mitochondrial Pi will not increase. We estimated mitochondrial Pi in this condition where excess Pi was externally supplemented. Notably, cells grown in high Pi had decreased mitochondrial Pi (Figure S5C). Furthermore, directly adding Pi to cells growing in high glucose decreased basal OCR (Figure S5D),

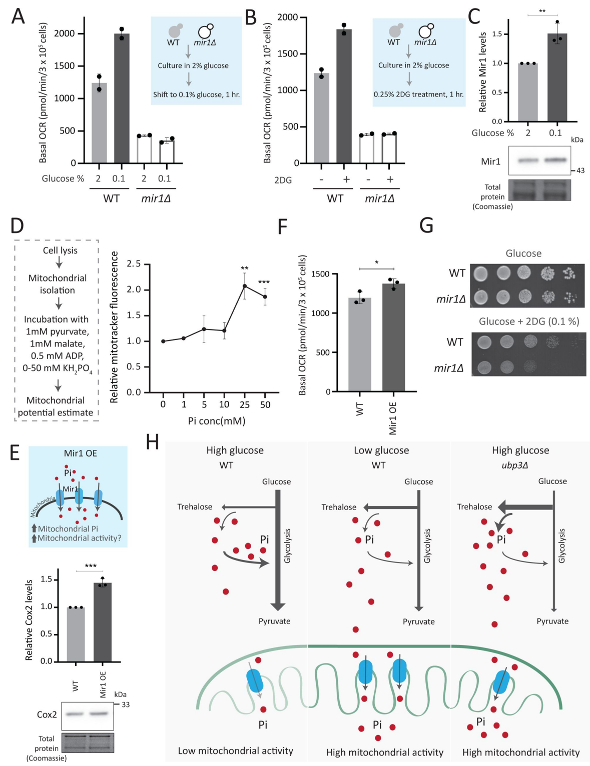

Figure 5

Mitochondrial Pi availability constrains mitochondrial activity under high glucose

A) Relationship of mitochondrial Pi transport and respiration after glucose removal. WT and *mir1Δ* cells were cultured in high (2%) glucose and shifted to low (0.1%) glucose for 1 hour. The normalized basal OCR, from two independent experiments (n=2) are shown. Data represent mean ± SD.
B) Requirement of mitochondrial Pi transport for switch to respiration upon glycolytic inhibition by 2DG. WT and *mir1Δ* cells were cultured in high glucose and treated with or without 0.25% 2DG for 1 hour. Basal OCR was measured from two independent experiments (n=2). Data represent mean ± SD. Also see figure S5A.
C) Glucose-dependent regulation of Mir1. Cells (with Mir1-HA) were grown in high glucose and shifted to low glucose (0.1% glucose) for 1 hour, and Mir1 levels compared. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD. Also see figure S5B.
D) Increasing Pi concentrations and mitochondrial activity in isolated mitochondria. Mitochondria were isolated from WT cells grown in high glucose, incubated with 1 mM pyruvate, 1 mM malate, 0.5 mM ADP and 0-50 mM KH_2PO_4. The mitochondrial activity was estimated by mitotracker fluorescence intensity, and intensities relative to the sample with 0 mM KH_2PO_4 is shown. Data represent mean ± SD from three biological replicates (n=3).
E) Effect of overexpressing Mir1 on Cox2 protein. WT (containing empty vector) and Mir1 overexpressing (Mir1OE) cells were grown in high glucose and Cox2 levels were estimated. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD. Also see figure S5F.
F) Effect of overexpressing Mir1 on basal OCR. The basal OCR in WT (containing empty vector) and Mir1OE in high glucose was measured from three independent experiments (n=3). Data represent mean ± SD.
G) Mitochondrial Pi transport requirement for growth after 2DG treatment. Shown are serial dilution growth assays in high glucose in the presence and absence of 0.1% 2DG, using WT and *mir1Δ* cells. The results after 40hrs incubation/30°C are shown.
H) A model illustrating how mitochondrial Pi availability controls mitochondrial activity. In high glucose, the decreased Pi due to high Pi consumption in glycolysis, along with the glucose-mediated repression of mitochondrial Pi transporters, decreases mitochondrial Pi availability. This reduces mitochondrial activity. In low glucose, increased mitochondrial Pi transporters and lower glycolytic flux increases mitochondrial Pi, leading to enhanced mitochondrial activity. In *ubp3Δ* cells in high glucose, high trehalose synthesis and lower glycolytic flux results in an increase in Pi. This increases mitochondrial Pi availability and thereby the mitochondrial activity.
Data information: *p<0.05, **p<0.01, ***p<0.001.

consistent with decreased mitochondrial Pi. These data indicate that in high glucose, simply supplementing Pi will not increase Pi access to the mitochondria. We therefore now asked, if we inhibit glycolytic flux and then supplement Pi, what would happen to mitochondrial activity. For this, we treated cells with 2DG, and subsequently added Pi and measured the OCR (Figure S5E). In this case, supplementing Pi increased the basal OCR (Figure S5E). Collectively, these data suggest that a combination of decreasing glycolysis and increasing Pi can together increase respiration. Next, in order to directly test mitochondrial activation based on external Pi availability, we isolated mitochondria, and estimated activity *in vitro* upon adding increasing Pi. Mitochondrial activity increased with increased Pi, with maximum activity observed with 25 mM Pi supplemented (**Figure 5D**). In a complementary experiment, we overexpressed the Mir1 transporter in wild-type cells, to increase Pi within mitochondria (Figure S5F). Mir1-OE cells have higher Cox2 levels and basal OCR (**Figure 5E, 5F**). Therefore, increasing Pi transport to mitochondria is sufficient to increase mitochondrial activity in high glucose.

Mitochondrial pyruvate transport is also required for mitochondrial respiration (45). We asked where Pi availability stands in a hierarchy of constraints for mitochondrial derepression, as compared to mitochondrial pyruvate transport. We measured the amounts of the Mpc3 subunit of the mitochondrial pyruvate carrier (MPC) complex (45, 46). Mpc3 protein increases in *ubp3Δ* (Figure S5G), correlating with the increased mitochondrial activity. Interestingly, in *ubp3Δ* grown in low Pi, Mpc3 further increased (Figure S5G), but as shown earlier this condition cannot increase OCR or mitochondrial activity (**Figure 4F**, S4F). Basal Mpc3 levels decrease in *mir1Δ*, but upon shifting to 0.1% glucose, Mpc3 increases in both WT and *mir1Δ*, with higher levels in *mir1Δ* (Figure S5H). Therefore, even where Mpc3 is high (*ubp3Δ* in low Pi, and *mir1Δ* in low glucose), mitochondrial activity remains low if Pi is restricted (**Figure 4F, 4H**). There was also no decrease in basal OCR in *mpc3Δ* in high glucose, and the basal OCR increased to the same level as of WT after shifting to 0.1% glucose (Figure S5I). Since Mpc3 changes with mitochondrial Pi availability (Figure S5G, S5H), we also measured Mpc3 in the Mir1OE. No further changes in Mpc3 were observed in Mir1OE (Figure S5J), indicating that increasing mitochondrial Pi alone need not increase Mpc3. Overall, although Mpc3 levels corelate with decreased glycolysis (*ubp3Δ* -Figure S5G, *ubp3Δ* in low Pi-Figure S5G, low glucose -Figure S5H), increased Mpc3 alone cannot increase mitochondrial activity and respiration in the absence of adequate mitochondrial Pi.

Finally, we asked how important mitochondrial Pi transport was for growth, under 2DG treatment. In the presence of 2DG, WT cells have only slightly decreased growth. In contrast, *mir1Δ* show a severe growth defect upon 2DG treatment (**Figure 5G**), revealing a synergetic effect of combining 2DG with inhibiting mitochondrial Pi transport. Therefore, the combined inhibition of glycolysis and mitochondrial Pi transport restricts the growth of glycolytic cells.

Collectively, mitochondrial Pi availability constrains glucose-mediated mitochondrial repression. Increasing available pools of Pi to enter the mitochondria is sufficient to induce mitochondrial activity.

Discussion

In this study, we highlight a role for Pi budgeting between cytosolic glycolysis and mitochondrial processes (which compete for Pi) in constraining mitochondrial repression (**Figure 5H**). Ubp3 controls this process (**Figure 1**), by maintaining the amounts of the glycolytic enzymes Pfk1 and GAPDH (Tdh2 and Tdh3) and thereby allows high glycolytic flux. At high fermentation rates, glycolytic enzymes levels at maximal activity maintain high glycolytic rates (effectively following zero-order kinetics), and therefore, changes in the enzyme levels will have a direct, proportionate effect on flux (47). The loss of Ubp3 decreases glycolytic flux, resulting in a systems-level, mass-action based rewiring of glucose metabolism where more G6P is routed towards trehalose synthesis and PPP (**Figure 2**, **Figure 5H**). Indeed, inhibiting just phosphofructokinase can reroute glucose flux from glycolysis to PPP (48–50), and our study now permits contextualized interpretations of these results. Such reallocations of glucose flux will collectively increase overall Pi, coming from the combined effect of increased Pi release from trehalose synthesis, and decreased Pi consumption via reduced GAPDH. This altered intracellular Pi economy increases Pi pools available to mitochondria, and increasing mitochondrial Pi is necessary and sufficient to increase respiration in high glucose (**Figure 4**, **Figure 5**). Mitochondrial Pi transport maintains mitochondrial activity in high glucose, and increases mitochondrial activity in low glucose (**Figure 4**, **Figure 5**). Finally, the mitochondrial Pi transporter Mir1 itself decreases in high glucose (**Figure 5**). Therefore, this glucose-dependent repression of Mir1 also restricts mitochondrial Pi availability (and thereby activity) in high glucose.

Traditionally, loss-of-function mutants of metabolic enzymes are used to understand metabolic state regulation. This approach negates nuanced investigations, since metabolic enzymes are often essential for viability. Furthermore, metabolic pathways have multiple, contextually regulated nodes, through which cells maintain their metabolic state. Therefore, alternate approaches to identify global regulators of metabolic states (as opposed to single enzymes), might uncover ways via which multiple nodes are simultaneously tuned, and can reveal unanticipated systems-level principles of metabolic state rewiring. In this study of glucose-mediated mitochondrial repression, the loss of the DUB Ubp3 decreases glycolytic flux by reducing the enzymes at two critical nodes in the pathway - Pfk1 and GAPDH. Unlike loss of function mutants, a reduction in amounts will only rewire metabolic flux. By 'hitting' multiple steps in glycolysis simultaneously, *ubp3Δ* have decreased Pi consumption, as well as increased Pi release. Such a cumulative phenomenon reveals more than inhibiting only GAPDH, where increased Pi comes only from reduced Pi consumption, and not from increased trehalose biosynthesis. Our serendipitous identification of a regulator which regulates multiple steps in glucose metabolism to change the metabolic environment, now suggests a general basis of mitochondrial regulation that would have otherwise remained hidden. Separately, finding the substrates of Ubp3 and whether Ubp3 directly regulates glycolytic enzymes are exciting future research questions requiring concurrent innovations in accessible chemical-biological approaches to study DUBs.

Because phosphates are ubiquitous, it is challenging to identify hierarchies of Pi-dependent processes in metabolic state regulation (33). Phosphate transfer reactions are the foundation of metabolism, driving multiple, thermodynamically unfavorable reactions (51, 52). Contextually, the laws of mass action predict that the relative rates of these reactions will regulate overall Pi balance, and contrarily the Pi allocation to Pi-dependent reactions will determine reaction rates (33, 35). Additionally, cells might control Pi allocations for different reactions via compartmentalizing Pi in organelles, to spatially restrict Pi availability (53–55). Our data collectively suggest a paradigm where the combination of factors regulates mitochondrial Pi and thereby activity. In glycolytic yeast cells growing in high glucose, mitochondrial Pi availability becomes restricted due to higher utilization of Pi in glycolysis compared to mitochondria. Consistent with this, a rapid decrease in Pi upon glucose addition has been observed (18, 19, 37). Interesting, *in vitro* studies with isolated mitochondria from tumor cells also find that decreasing Pi levels decreases respiration (19), which would be consistent with this scenario. Further, supplementing Pi correlates with decreased mitochondrial repression in tumors (18, 56). By increasing Pi through a systems-level rewiring of glucose metabolism (such as in *ubp3Δ* cells), cells can collectively increase mitochondrial access to Pi. This Pi budgeting determines mitochondrial activity. Supplementing Pi under conditions of low glycolysis (where mitochondrial Pi transport is enhanced), as well as directly supplementing Pi to isolated mitochondria, increases respiration (**Figure 5**, S5). Therefore, in order to deprepress mitochondria, a combination of increased Pi along with decreased glycolysis is required. An additional systems-level phenomenon that might regulate Pi transport to the mitochondria is the decrease in cytosolic pH upon decreased glycolysis (57, 58). Since mitochondrial Pi transport itself dependeds on the proton gradient, a low cytosolic pH would favour mitochondrial Pi transport (59). Therefore, decreasing glycolysis (2DG treatment, deletion of Ubp3, and decreased GAPDH activity), and increasing cytosolic Pi might synergestically increase mitochondria Pi transport, and increase respiration. Alternately, increasing mitochondrial Pi transporter amounts can achieve the same result, as seen by overexpressing Mir1 (**Figure 5**). A similar observation has been reported in *Arabidopsis*, reiterating an evolutionarily conserved role for mitochondrial Pi in controlling respiration (60). Relatedly, glycolytic inhibition can suppress cell proliferation in Warburg-positive tumors (61, 62), or inflammatory responses (63). However, these cells survive by switching to mitochondrial respiration (64, 65), requiring alternate approaches to prevent their proliferation (66). Inhibiting mitochondrial Pi transport in combination with glycolytic inhibition could restrict the proliferation of Warburg/Crabtree positive cells.

Since its discovery in the 1920s, the phenomenon of accelerated glycolysis with concurrent mitochondrial repression has been intensely researched. Yet, the biochemical constraints for glucose-mediated mitochondrial repression remains unresolved. One hypothesis suggests that the availability of glycolytic intermediates might determine the extent of mitochondrial repression. F1,6BP inhibits complex III and IV of the electron transport chain (ETC) in Crabtree positive yeast (14–17). Similarly, the ratio between G6P and F1,6BP regulates the extent of mitochondrial repression (16, 17). Although G6P/F6P accumulates in *ubp3Δ* (**Figure 2C**), this is not the case in *tdh2Δtdh3Δ* (GAPDH mutant) (Figure S3D), suggesting that G6P/F6P accumulation in itself is not the criterion to increase mitochondrial activity. Separately, the competition for common metabolites/co-factors between glycolysis and respiration (such as ADP, Pi or pyruvate) could drive this phenomenon (14, 18). Here we observe that Mpc3 mediated mitochondrial pyruvate transport alone cannot increase respiration. An additional consideration is the possible contribution of changes in ADP in regulating mitochondrial activity, where the use of ADP in glycolysis might limit mitochondrial ADP. Therefore, when Pi changes as a consequence of glycolysis, it coud be imagined that a change in ADP balance can coincidentally occur. However, prior studies show that even though cytosolic ADP decreases in the presence of glucose, this does not limit mitochondrial ADP uptake, or decrease respiration, due to the very high affinity of the mitochondrial ADP transporter (14, 19). These collectively reiterate the importance of Pi access and transport to mitochondria in constraining mitochondrial respiration. Indeed, this interpretation can also contextually explain observations from other model systems where mitochondrial Pi transport seems to regulate respiration (67, 68).

We parsimoniously suggest that Pi access to the mitochondria as a key constraint for mitochondrial repression under high glucose. In a hypothetical scenario, a single-step event in evolution, reducing mitochondrial Pi transporter amounts, and/or increasing glycolytic flux (to deplete cytosolic Pi), will result in whole-scale metabolic rewiring to repress mitochondria. More elaborate regulatory events can easily be imagined as subsequent adaptations to enforce mitochondrial repression. Given the central role played by Pi, something as fundamental as access to Pi will constrain mitochondrial repression. Concurrently, the rapid incorporation of Pi into faster glycolysis can give cells a competitive advantage, while also sequestering Pi in the form of usable ATP. Over the course of evolution, this could conceivably drive other regulatory mechanisms to enforce mitochondrial repression, leading to the currently observed complex regulatory networks and signaling programs observed in the Crabtree effect, and other examples of glucose-dependent mitochondrial repression.

Materials and methods

Statistics and graphing

Unless otherwise indicated, statistical significance for all indicated experiments were calculated using unpaired Student's t-tests (GraphPad prism 9.0.1). Graphs were plotted using GraphPad prism 9.0.1

Yeast strains, media, and growth conditions

A prototrophic CEN.PK strain of *Saccharomyces cerevisiae* (WT) (69) was used. Strains are listed in appendix table S1. Gene deletions, chromosomal C-terminal tagged strains were generated by PCR mediated gene deletion/tagging (70). Media compositions, growth conditions and CRISPR-Cas9 based mutagenesis are described in extended methods (SI appendix).

Mitotracker fluorescence

Mitotracker fluorescence was measured using Thermo Varioscan^{TM} LUX multimode plate reader (579/599 excitation/emission). Detailed protocol is described in extended methods. Mitotracker fluorescence were normalized using OD600 of each sample and relative fluorescence intensity calculated.

Protein extraction and Western blotting

Total protein was precipitated, extracted using trichloroacetic acid (TCA) as described earlier (71). Blots were quantified using ImageJ software. Detailed protocol is described in extended methods.

Basal OCR measurement

The basal OCR was measured using Agilent Seahorse XFe24 analyzer. Basal OCR readings were normalized for cell number (using OD_{600} of samples) in each well. The detailed methods are described in extended methods.

RNA extraction and RT-qPCR

The RNA extraction was done using the hot phenol extraction method as described in (71). The isolated RNA was DNase treated, and used for cDNA synthesis. Superscript III reverse transcriptase enzyme (Invitrogen) was used for cDNA synthesis and RT-qPCR was performed using KAPA SYBR FAST qRT PCR kit (KK4602, KAPA Biosystems). Taf10 was used as a control for normalisation and the fold change in mRNA levels were calculated by 2^-ddct method.

ATP, ethanol and Pi measurements

ATP levels were measured by ATP estimation kit (Thermo Fisher A22066). Ethanol concentration in the medium was estimated using potassium dichromate based assay described in (72) with modifications. Pi was estimated using a malachite green phosphate assay kit (Cayman chemicals, 10009325). Detailed sample collection and assay protocols are described in extended methods.

Metabolite extraction and analysis by LC-MS/MS

The steady state levels and relative ^{13}C label incorporation into metabolites were estimated by quantitative LC-MS/MS methods as described in (73). Detailed methodology is extensively described in extended methods. Peak area measurements are listed in supplement file 1.

Mitochondrial isolation

Mitochondria was isolated by immunoprecipitation as described in (74, 75) with modifications. The detailed protocol is described in extended methods (SI appendix). The eluted mitochondria were used in malachite green assay for Pi estimation, boiled with SDS-glycerol buffer for western blots or incubated with mitotracker CMXROS with mitochondria-activation buffer for mitotracker assays.

Data availability

The LC-MS/MS peak area for all the metabolites can be found in supplement file 1. This study includes no data deposited in external repositories.

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Reviewer #1 (Public Review):

The study by Vengayil et al. presented a role for Ubp3 for mediating inorganic phosphate (Pi) compartmentalization in cytosol and mitochondria, which regulates metabolic flux between cytosolic glycolysis and mitochondrial processes. Although the exact function of increased Pi in mitochondria is not investigated, findings have valuable implications for understanding the metabolic interplay between glycolysis and respiration under glucose-rich conditions. They showed that UBP3 KO cells regulated decreased glycolytic flux by reducing the key Pi-dependent-glycolytic enzyme abundances, consequently increasing Pi compartmentalization to mitochondria. Increased mitochondria Pi increases oxygen consumption and mitochondrial membrane potential, indicative of increased oxidative phosphorylation. In conclusion, the authors reported that the Pi utilization by cytosolic glycolytic enzymes is a key process for mitochondrial repression under glucose conditions.

However, the main claims are only partially supported by the low number of repeats and utilizing only one strain background, which decreased the overall rigor of the study. The full-power yeast model could be utilized with testing findings in different backgrounds with increased biological repeats in many assays described in this study. In the yeast model, it has been well established that many phenotypes are genotype/strain dependent (Liti 2019, Gallone 2016, Boekout 2021, etc...). with some strains utilizing mitochondrial respiration even under high glucose conditions (Kaya 2021). It would be conclusive to test whether wild strains with increased respiration under high glucose conditions would also be characterized by increased mitochondrial Pi.

It is not described whether the drop in glycolytic flux also affects TCA cycle flux. Are there any changes in the pyruvate level? If the TCA cycle is also impaired, what drives increased mitochondrial respiration?

In addition, some of the important literature was also missed in citation and discussion. For example, in a recent study (Ouyang et al., 2022), it was reported that phosphate starvation increases mitochondrial membrane potential independent of respiration in yeast and mammalian cells, and some of the conflicting results were presented in this study.

An additional experiment with strains lacking mitochondrial DNA under phosphate-rich and restricted conditions would further strengthen the result.

Western blot control panels should include entire membrane exposure, and non-cut western blots should be submitted as supplementary.

In Figure 4, it is shown that Pi addition decreases basal OCR to the WT level. However, the Cox2 level remains significantly higher. This data is confusing as to whether mitochondrial Pi directly regulates respiration or not.

Representative images of Ubx3 KO and wild-type strains stained with CMXRos are missing.

Overall, mitochondrial copy number and mtDNA copy number should be analyzed in WT and Ubo3 KO cells as well as Pi-treated and non-treated cells, and basal OCR data should be normalized accordingly. The reported normalization against OD is not appropriate.

Reviewer #2 (Public Review):

Summary:
Cells cultured in high glucose tend to repress mitochondrial biogenesis and activity, a prevailing phenotype type called Crabree effect that is observed in different cell types and cancer. Many signaling pathways have been put forward to explain this effect. Vengayil et al proposed a new mechanism involved in Ubp3/Ubp10 and phosphate that controls the glucose repression of mitochondria. The central hypothesis is that Δubp3 shifts the glycolysis to trehalose synthesis, therefore leading to the increase of Pi availability in the cytosol, then mitochondria receive more Pi, and therefore the glucose repression is reduced.

Strengths:
The strength is that the authors used an array of different assays to test their hypothesis. Most assays were well-designed and controlled.

Weaknesses:
I think the main conclusions are not strongly supported by the current dataset.

1. Although the authors discovered Δubp3 cells have higher Pi and mitochondrial activity than WT in high glucose, it is not known if WT cultured in different glucose concentration also change Pi that correlate with the mitochondrial activity. The focus of the research on Δubp3 is somewhat artificial because Δubp3 not only affects glycolysis and mitochondria, but many other cellular pathways are also changed. There is no idea whether culturing cells in low glucose, which de-repress the mitochondrial activity, involves Ubp3 or not. Similarly, the shift of glycolysis to trehalose synthesis is also not relevant to the WT cells cultured in a low-glucose situation.

2. The central hypothesis that Pi is the key constraint behind the glucose repression of mitochondrial biogenesis/activity is supported by the data that limiting Pi will suppress mitochondrial activity increase in these conditions (e.g., Δubp3). However, increasing the Pi supply failed to increase mitochondrial activity. The explanation put forward by the authors is that increased Pi supply will increase glycolysis activity, and somehow even reduce the

mitochondrial Pi. I cannot understand why only the increased Pi supply in Δubp3, but not the increased Pi by medium supplement, can increase mitochondrial activity. The authors said "...that ubp3Δ do not increase mitochondrial Pi by merely increasing the Pi transporters, but rather by increasing available Pi pools". They showed that Δubp3 mitochondria had higher Pi but WT cells with medium Pi supplement showed lower Pi, it is hard to understand why the same Pi increase in the cytosol had a different outcome in mitochondrial Pi. Later on, they showed that the isolated mito exposed to higher Pi showed increased activity, so why can't increased Pi in intact cells increase mito activity? Moreover, they first showed that Δubp3 had a Mir1 increase in Fig3A, then showed no changes in FigS4G. It is very confusing.

3. Given that there is no degradation difference for these glycolytic enzymes in Δubp3, and the authors found transcriptional level changes, suggests an alternative possibility where Δubp3 may signal through unknown mechanisms to parallelly regulate both mitochondrial biogenesis and glycolytic enzyme expression. The increase of trehalose synthesis usually happens in cells under proteostasis stress, so it is important to rule out whether Δubp3 signals these metabolic changes via proteostasis dysregulation. This echoes my first point that it is unknown whether wild-type cells use a similar mechanism as Δubp3 cells to regulate the glucose repression of mitochondria.

4. Other major concerns:
a. The authors selectively showed a few proteins in their manuscript to support their conclusion. For example, only Cox2 and Tom70 were used to illustrate mitochondrial biogenesis difference in line 97. Later on, they re-analyzed the previous MS dataset from Isasa et al 2015 and showed a few proteins in Fig3A to support their conclusion that Δubp3 increases mitochondrial OXPHOS proteins. However, I checked that MS dataset myself and saw that many key OXPHOS proteins do not change, for example, both ATP1 and ATP2 do not change, which encode the alpha and beta subunits of F1 ATPase. They selectively reported the proteins' change in the direction along with their hypothesis.
b. The authors said they deleted ETC component Cox2 in line 111. I checked their method and table S1, I cannot figure out how they selectively deleted COX2 from mtDNA. This must be a mistake.
c. They used sodium azide in a lot of assays to inhibit complex IV. However, this chemical is nonspecific and broadly affects many ATPases as well. Not sure why they do not use more specific inhibitors that are commonly used to assay OCR in seahorse.
d. The authors measured cellular Pi level by grinding the entire cells to release Pi. However, this will lead to a mix of cytosolic and vacuolar Pi. Related to this caveat, the cytosol has ~50mM Pi, while only 1-2mM of these glycolysis metabolites, I am not sure why the reduction of several glycolysis enzymes will cause significant changes in cytosolic Pi levels and make Pi the limiting factor for mitochondrial respiration. One possibility is that the observed cytosolic Pi level changes were caused by the measurement issue mentioned above.
e. The authors used Δmir1 and MIR1 OE to show that Pi viability in the mitochondrial matrix is important for mitochondrial activity and biogenesis. This is not surprising as Pi is a key substrate required for OXPHOS activity. I doubt the approach of adding a control to determine whether Pi has a specific regulatory function, while other OXPHOS substrates, like ADP, O2 etc do not have the same effect.